

pgAdmin 4

File Object Tools Edit View Window Help

Object Explorer

- Servers (2)
 - PostgreSQL 13
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 - college_db
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 - Tablespaces
 - PostgreSQL 17

Query

```
1 explain
2 select s.first_name,
3        c.course_name
4 from students s
5 join enrollments e
6 on s.student_id = e.student_id
7 join courses c
8 on e.course_id = c.course_id;
```

Query History

Data Output Messages Notifications

Showing rows: 1 to 9 Page No: 1 of 1

QUERY PLAN

1	Hash Join (cost=27.88..60.14 rows=1450 width=436)
2	Hash Cond: (e.course_id = c.course_id)
3	-> Hash Join (cost=13.60..41.98 rows=1450 width=122)
4	Hash Cond: (e.student_id = s.student_id)
5	-> Seq Scan on enrollments e (cost=0.00..24.50 rows=1450 width=...
6	-> Hash (cost=11.60..11.60 rows=160 width=122)
7	-> Seq Scan on students s (cost=0.00..11.60 rows=160 width=...
8	-> Hash (cost=11.60..11.60 rows=160 width=122)

Total rows: 9 Query complete 00:00:00.157

CRLF Ln 8, Col 30

20:16 01-07-2026

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Query

```
on courses(course_code);
create index idx_enrollments_student
on enrollments(student_id);
select indexname,
       tablename
from pg_indexes
where schemaname = 'public';
```

Query History

Data Output Messages Notifications

	indexname	tablename
	name	name
5	courses_course_code...	courses
6	enrollments_pkey	enrollments
7	professors_pkey	professors
8	professors_email_key	professors
9	idx_students_departm...	students
10	idx_courses_code	courses
11	idx_enrollments_stud...	enrollments

Total rows: 11 Query complete 00:00:00.116

CRLF Ln 22, Col 29

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Query

```
23 explain
24 select s.first_name,
25        c.course_name
26 from students s
27 join enrollments e
28 on s.student_id = e.student_id
29 join courses c
30 on e.course_id = c.course_id;
```

Query History

Data Output Messages Notifications

Showing rows: 1 to 9 Page No: 1 of 1

	QUERY PLAN
1	Hash Join (cost=2.20..3.29 rows=4 width=436)
2	Hash Cond: (e.course_id = c.course_id)
3	-> Hash Join (cost=1.11..2.17 rows=4 width=122)
4	Hash Cond: (e.student_id = s.student_id)
5	-> Seq Scan on enrollments e (cost=0.00..1.04 rows=4 width=8)
6	-> Hash (cost=1.05..1.05 rows=5 width=122)
7	-> Seq Scan on students s (cost=0.00..1.05 rows=5 width=...
8	-> Hash (cost=1.04..1.04 rows=4 width=122)

Total rows: 9 Query complete 00:00:00.099

CRLF Ln 29, Col 31

20:18 01-07-2026